STPM Physics · Semester 3 · Chapter 21

WAVE MOTION

21.1 Progressive Waves

DEFINITIONS & LAWS

- A **progressive (travelling) wave** is a wave whose profile moves along the medium; it transfers energy from one point to another through a medium (or vacuum) by oscillations of the particles (or fields), without transferring matter.
- **Wavelength** is the distance between two successive points that are in phase on the wave (the distance travelled by the wave in one period).
- A **wavefront** is the locus of points which vibrate in phase.
- **Amplitude** is the maximum displacement of a particle from its equilibrium position; **period** is the time for one complete oscillation of a particle.
- **Wave speed** is the distance travelled by the wavefront per unit time.
- **Phase difference** between two points on a progressive wave depends on their separation along the direction of travel.
- In a transverse wave the oscillations are perpendicular to the direction of propagation; in a longitudinal wave they are parallel to it.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Wave speed	$v = f\lambda$	v = wave speed (m s^{-1}); f = frequency (Hz); λ = wavelength (m)
Wave travelling in +x direction	$y = A\sin(\omega t - kx)$	y = displacement (m); A = amplitude (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad m^{-1}); $\omega = 2\pi f$ (rad s^{-1})
Wave speed in terms of ω and k	$v = \frac{\omega}{k} = \frac{\lambda}{T}; f = \frac{\omega}{2\pi}$	v = wave speed (m s^{-1}); ω = angular frequency (rad s^{-1}); k = wave number (rad m^{-1}); T = period (s); f = frequency (Hz)
Phase difference	$\varphi = \frac{2\pi}{\lambda}\Delta x; \varphi = \frac{2\pi\Delta t}{T}$	φ = phase difference (rad); Δx = path separation (m); Δt = time difference (s); T = period (s)
Wave travelling in −x direction	$y = A\sin(\omega t + kx)$	Travels in the −x direction; symbols as above

TIPS TO ANSWER

- Read ω and k from the equation by comparison with $y = A\sin(\omega t - kx)$; then $v = \frac{\omega}{k}$ and $f = \frac{\omega}{2\pi}$.
- Phase difference: write $\varphi = \frac{2\pi\Delta x}{\lambda}$, substitute, and give the answer in radian (and degree if asked) with 3 s.f.
- For graphs: y against x gives λ and A; y against t gives T and A. Label axes with units.
- Say that energy is transferred in the direction of wave propagation but the particles only oscillate about their equilibrium positions.
- When asked to write a wave equation, evaluate $k = \frac{2\pi}{\lambda}$ and $\omega = 2\pi f$ as numbers. Do not leave the answer in terms of π unless the question asks for it, or the answer mark is lost. Example: λ = 10 cm, f = 400 Hz, A = 2.0 cm gives $y = 0.020\sin(63x - 2.5 \times 10^3 t)$ m.
- To find the phase angle φ from a displacement graph, read y at t = 0 (or x = 0): $y_0 = A\sin\varphi$. For example, $y_0 = 0.50A$ gives $\varphi = 30° = \frac{\pi}{6}$ rad.

SECTION A TRAPS

Trap: Using $v = \omega k$ instead of $\frac{\omega}{k}$. Check units: $\frac{\omega}{k}$ gives m s^{-1}.

Trap: Taking the number multiplying t as f. In $y = A\sin(50t - 4.0x)$, 50 is ω (rad s^{-1}), so $f = \frac{50}{2\pi} \approx 8.0$ Hz.

Trap: Phase difference without 2π: $\varphi = \frac{\Delta x}{\lambda}$ is the fraction of a cycle, not the phase angle in radian.

Trap: Reading $y = A\sin[2\pi(\frac{t}{T} - \frac{x}{\lambda})]$ with the denominators swapped; $\frac{t}{T}$ gives period and $\frac{x}{\lambda}$ gives wavelength.

TRICKS

Trick: Two points separated by one wavelength are in phase (2π); separated by $\frac{\lambda}{2}$ they are in antiphase (π); by $\frac{\lambda}{4}$ the phase difference is $\frac{\pi}{2}$.

Trick: In the form $\sin[2\pi(\frac{t}{T} - \frac{x}{\lambda})]$, read T and λ directly and take $v = \frac{\lambda}{T}$.

21.2 Wave Intensity

DEFINITIONS & LAWS

- **Intensity** of a wave is the rate of energy transferred per unit area perpendicular to the direction of propagation (power per unit area), unit W m^{-2}.
- Intensity is directly proportional to the square of the amplitude (for the same frequency and medium).
- For a point source radiating uniformly in all directions with no absorption, intensity is inversely proportional to the square of the distance from the source (inverse square law).
- Explain the inverse square law: the same power spreads over the surface of a sphere of area $4\pi r^2$, which increases as r^2.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Intensity	$I = \frac{P}{A}$	I = intensity (W m^{-2}); P = power (W); A = area normal to the wave (m^2)
Intensity of a point source	$I = \frac{P}{4\pi r^2}$	Point source; r = distance from the source (m); P = power (W)
Intensity and amplitude	$I \propto A^2$; $\frac{I_2}{I_1} = (\frac{A_2}{A_1})^2$	A = amplitude; I = intensity (W m^{-2})
Inverse square law	$I_1 r_1^2 = I_2 r_2^2$; $I_2 = I_1(\frac{r_1}{r_2})^2$	I_1, I_2 = intensities at distances r_1, r_2 from the source (W m^{-2}, m)
Amplitude of a spherical wave	$A \propto \frac{1}{r}$	Spherical wave; A = amplitude; r = distance from the source (m)

TIPS TO ANSWER

- Choose between $I \propto \frac{1}{r^2}$ (distance change) and $I \propto A^2$ (amplitude change), and write the relation before substituting.
- Show the ratio: $I_2 = I_1(\frac{r_1}{r_2})^2$ with values and unit, then the answer to 3 s.f. in W m^{-2}.
- State the assumptions: point source, spreads uniformly, no absorption by the medium.
- For power calculations, use area $4\pi r^2$ and give the answer with unit W m^{-2}.

SECTION A TRAPS

Trap: Using $I \propto \frac{1}{r}$ instead of $\frac{1}{r^2}$. Doubling the distance gives one quarter, not one half.

Trap: Inverting the ratio: moving farther must decrease the intensity, so the factor $(\frac{r_1}{r_2})^2$ is less than 1.

Trap: Tripling the amplitude triples the intensity. Wrong: $I \propto A^2$, so the intensity becomes 9 times.

Trap: Using the diameter as r in $I = \frac{P}{4\pi r^2}$.

TRICKS

Trick: Distance × 3 → intensity ÷ 9; amplitude × 3 → intensity × 9.

Trick: Intensity down by a factor of 4 means distance up by a factor of 2: $r_2 = r_1\sqrt{\frac{I_1}{I_2}}$.

21.3 Principle of Superposition

DEFINITIONS & LAWS

- **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.
- **Coherent sources** are sources that emit waves of the same frequency with a constant phase difference.
- **Constructive interference** occurs when the waves meet in phase: the path difference is nλ (n = 0, 1, 2, ...), giving the maximum resultant amplitude.
- **Destructive interference** occurs when the waves meet in antiphase: the path difference is (n + ½)λ, giving the minimum resultant amplitude.
- A stable, observable interference pattern requires coherent waves of similar amplitude and the same frequency.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Constructive interference	$\Delta x = n\lambda$	Δx = path difference (m); λ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources
Destructive interference	$\Delta x = (n + ½)\lambda$	Δx = path difference (m); λ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources
Phase difference	$\varphi = \frac{2\pi}{\lambda}\Delta x$	φ = phase difference (rad); Δx = path difference (m); λ = wavelength (m)

Quantity / law	Formula	Symbols, units & conditions
Resultant amplitude	$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos\varphi$	Two waves of amplitudes A_1 and A_2 (m); φ = phase difference (rad)
Intensity of equal waves	$I = 4I_0\cos^2\frac{\varphi}{2}$	Equal waves; I_0 = intensity of each wave (W m^{-2}); φ = phase difference (rad)
Special cases of phase difference	$\varphi = \frac{\pi}{2}$: $A = \sqrt{A_1^2 + A_2^2}$; $\varphi = 0$: $A = A_1 + A_2$; $\varphi = \pi$: $A = \lvert A_1 - A_2 \rvert$	A = resultant amplitude (m); A_1, A_2 = amplitudes of the two waves (m); φ in rad

TIPS TO ANSWER

- Compute $\frac{\Delta x}{\lambda}$ and state the result: a whole number gives constructive; an odd number of half-wavelengths gives destructive.
- Define coherence completely: same frequency and constant phase difference (not just 'same amplitude').
- For the amplitude of the resultant wave, state φ, then substitute into the amplitude relation with unit.
- Compare the intensity with I_0 using $I \propto A^2$, and state the answer in terms of I_0.
- For two waves of equal amplitude A with phase difference φ, find the resultant amplitude with the cosine rule (vector triangle, as in Semester 1): $A'^2 = A^2 + A^2 - 2A^2\cos(180° - \varphi)$. Then use $I \propto A^2$: $\frac{I'}{I_0} = \frac{A'^2}{A^2}$. For $\varphi = 30°$, $A'^2 = 3.73A^2$, so $I' = 3.7I_0$ (not $2I_0$ or $4I_0$). This is within the syllabus.

SECTION A TRAPS

Trap: Concluding destructive interference just because the path difference is 'not an integer'. Check whether $\frac{\Delta x}{\lambda}$ is a half-integer, e.g. 1.5.

Trap: Simply adding amplitudes (3 + 4 = 7) for a phase difference of $\frac{\pi}{2}$; the correct result is $\sqrt{3^2 + 4^2} = 5$.

Trap: Converting the path difference to phase without 2π: $\varphi = \frac{2\pi\Delta x}{\lambda}$, so $\Delta x = \frac{\lambda}{3}$ gives $\frac{2\pi}{3}$ rad.

Trap: Saying that waves of different frequencies give a stable pattern; a constant phase difference requires the same frequency.

TRICKS

Trick: $\frac{\Delta x}{\lambda}$ = 3.0, 1.5, 0.75: integer → bright (constructive); half-integer → dark (destructive); anything else is in between.

Trick: $I = 4I_0\cos^2\frac{\varphi}{2}$: $\varphi = \frac{2\pi}{3}$ gives I_0; $\varphi = \pi$ gives 0; $\varphi = 0$ gives $4I_0$.

21.4 Standing Waves

DEFINITIONS & LAWS

- A **standing (stationary) wave** is a wave whose profile does not move; it is formed by the superposition of two progressive waves of the same frequency and amplitude travelling in opposite directions.
- A **node** is a point of zero amplitude (permanent destructive interference); an **antinode** is a point of maximum amplitude.
- Adjacent nodes (or adjacent antinodes) are separated by $\frac{\lambda}{2}$; a node and the adjacent antinode are separated by $\frac{\lambda}{4}$.
- Differences from a progressive wave: a standing wave does not transfer energy along the medium; its amplitude varies with position (zero at nodes); particles between two adjacent nodes oscillate in phase, and particles in adjacent segments oscillate in antiphase.
- For a string fixed at both ends there is a node at each end; only wavelengths with an integral number of half-wavelengths fit on the string.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Standing wave on string fixed at both ends	$L = \frac{n\lambda}{2}$; $f_n = \frac{nv}{2L}$	n = 1, 2, 3, …; v = wave speed on the string (m s^{-1}); L = length (m); λ = wavelength (m); f_n = frequency of the nth mode (Hz)
Fundamental and harmonics	$f_1 = \frac{v}{2L}$; $f_n = nf_1$	f_1 = fundamental frequency (Hz); f_n = frequency of the nth harmonic (Hz); v = wave speed (m s^{-1}); L = length (m)
Standing wave equation	$y = 2A\sin(kx)\cos(\omega t)$	$k = \frac{2\pi}{\lambda}$ (rad m^{-1}); A = amplitude of each travelling wave (m); ω = angular frequency (rad s^{-1}); nodes at kx = nπ
Node spacing	Distance between adjacent nodes = $\frac{\lambda}{2} = \frac{\pi}{k}$	λ = wavelength (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad m^{-1})

Quantity / law	Formula	Symbols, units & conditions
Wave speed on a string	$v = \sqrt{\frac{T}{\mu}}$	Stretched string; T = tension (N); μ = mass per unit length (kg m^{-1})

TIPS TO ANSWER

- Sketch the pattern with N at both ends for a string and label the nodes, antinodes and the wavelength on the diagram.
- State the condition $L = \frac{n\lambda}{2}$, then $f = \frac{v}{\lambda} = \frac{nv}{2L}$; give the answer to 3 s.f. in Hz.
- In an 'explain' question state: formed by superposition of the incident and reflected waves of the same frequency and amplitude travelling in opposite directions.
- From $y = 2A \sin(kx) \cos(\omega t)$ identify k, then $\lambda = \frac{2\pi}{k}$ and the node separation $\frac{\lambda}{2}$.
- Understand what "energy is not transferred" means: a standing wave stores energy in the vibrating medium. This is why standing waves are the sources of sound (stretched strings, closed and open pipes, drums).
- The standing wave on a stretched string is a **transverse** wave, but the sound wave it produces in air is **longitudinal**. Read whether the question asks about the source of sound or the sound wave.
- Practise drawing any harmonic quickly before the exam (e.g. the fifth harmonic of a closed pipe: $L = \frac{5\lambda}{4}$, three nodes and three antinodes along the pipe), so you do not lose time working it out in the exam.

SECTION A TRAPS

Trap: Taking the node separation to be λ rather than $\frac{\lambda}{2}$.

Trap: Using $f_1 = \frac{v}{L}$ for a string fixed at both ends; the fundamental has $L = \frac{\lambda}{2}$, so $f_1 = \frac{v}{2L}$.

Trap: Counting the gaps between nodes wrongly: from the 2nd node to the 5th node there are three gaps, i.e. $3 \times \frac{\lambda}{2}$.

Trap: Saying that a standing wave transports energy along the string, or that all points have the same amplitude.

TRICKS

Trick: The nth harmonic is n times the fundamental; the third harmonic of a 200 Hz fundamental is 600 Hz.

> **Trick:** From the number of gaps g between two nodes, distance = $g\frac{\lambda}{2}$; from a node at 0 and k, the spacing is $\frac{\pi}{k}$.

21.5 Electromagnetic Waves

DEFINITIONS & LAWS

- **Electromagnetic waves** are transverse waves consisting of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation; they do not need a material medium.
- All electromagnetic waves travel at the same speed, $c = 3.00\times10^{8}$ m s^{-1}, in vacuum; they differ in frequency and wavelength.
- In an electromagnetic wave, the electric and magnetic fields oscillate in phase.
- Order of the spectrum, from longest wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays.
- Electromagnetic waves can be polarised, reflected, refracted, diffracted and show interference (properties of transverse waves).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Wave equation for EM waves	$c = f\lambda$	c = speed of light in vacuum (3.00×10^{8} m s^{-1}); f = frequency (Hz); λ = wavelength (m)
Field amplitudes	$E_0 = cB_0$	E_0 = electric field amplitude (V m^{-1}); B_0 = magnetic field amplitude (T); c = speed of light (m s^{-1})
Speed of light	$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$	$\mu_0 = 4\pi\times10^{-7}$ H m^{-1}; $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
Photon energy	$E = hf = \frac{hc}{\lambda}$	E = photon energy (J); $h = 6.63\times10^{-34}$ J s; f = frequency (Hz); λ = wavelength (m)
Time of travel	$t = \frac{d}{c}$	t = time (s); d = distance (m); c = speed of light (m s^{-1})

TIPS TO ANSWER

- State the nature: transverse; E and B perpendicular to each other and to the direction of propagation; in phase.
- For calculations: formula → substitution with $c = 3.00\times10^{8}$ m s^{-1} → answer to 3 s.f. with unit and in standard form.

- For the spectrum, order by wavelength or frequency and state the direction, e.g. 'increasing frequency'.
- State that no medium is required, so EM waves travel through vacuum.

SECTION A TRAPS

Trap: Reversing the order of the spectrum: radio waves have the longest wavelength and lowest frequency; X-rays are shorter than ultraviolet.

Trap: Using B = cE instead of $B = \frac{E}{c}$, which gives an absurd 10^{11} T; B is of the order of 10^{-6} T for E of a few hundred V m^{-1}.

Trap: Saying that E and B are 90° out of phase. They are in phase (both reach maxima together).

Trap: Saying that EM waves are longitudinal or that they need a medium.

TRICKS

Trick: Quick check: $\lambda = \frac{c}{f} = 3\times\frac{10^8}{1.5}\times10^{14} = 2\times10^{-6}$ m; always examine the power of ten against the spectrum.

Trick: $\frac{E}{B}$ must equal 3×10^8 m s^{-1} in vacuum; use it to check magnitudes in options.